

The Holy Grail of Investing

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ABSTRACT. A 200-day trend filter on QQQ, rotating TQQQ into 1× QQQ in the bear leg, compounds \$10,000 into \$543,000 over 1999–2026 — 3.6× the terminal wealth of unleveraged QQQ buy-and-hold. The strategy sits in the lineage of Gayed & Bilello (2016, Charles H. Dow Award), Kane (2018), Bogleheads HEDGEFUNDIE (2019), and Masonson (2025), with the bear-leg asset choice as the variable under examination. Combinatorial purged cross-validation across 45 paths gives median CAGR 6.38% (IQR [2.39%, 16.18%]) versus 3.95% for QQQ buy-and-hold, with $P(\text{CAGR} > 15\%) = 31.1\%$ vs 8.9%. A parameter robustness grid of $\{\text{SMA}, \text{EMA}\} \times \{5, 10, 20, 50\} \times \{150, 200, 250\}$ ranks EMA(5)/EMA(200) sixth of 24 — not uniquely optimal, but comfortably above grid median. Deflated Sharpe analysis shows the observed edge survives against an *a priori* specification but fails under any realistic multiple-testing correction; the strategy is a regime-concentrated improvement over buy-and-hold, not a tested-across-universes discovery. Earlier published variants with T-bills in the bear leg deliver nearly identical Sharpe (0.545 vs 0.545) and marginally better Calmar (0.189 vs 0.166). The honest framing: this is a defensible implementation of a well-known strategy family, validated against adversarial statistical scrutiny, not a novel alpha source.

1. Context and Prior Literature

The strategy evaluated in this note is a variant of a well-established family. Before presenting any results, I locate the contribution within that lineage.

Gayed & Bilello (2016), “Leverage for the Long Run.” The canonical academic reference, winner of the 2016 Charles H. Dow Award. The rule: when S&P 500 closes above its 200-day simple moving average, hold leveraged S&P exposure (1.25×, 2×, or 3×); when below, rotate to Treasury bills. Reports roughly five rotations per year across a range of trend-filter lengths (10/20/50/100/200 SMA). The core mechanism — leverage is profitable only conditional on a favorable trend regime — is established here.

Kane (2018, Seeking Alpha) popularized the Gayed rule for retail use with UPRO and TQQQ explicitly. The substitution of EMA for SMA and faster-parameter crossovers became common in this ancestor.

Bogleheads “HEDGEFUNDIE's Excellent Adventure” (2019) is the most active retail forum for leveraged-ETF strategy discussion. The original UPRO/TMF risk-parity thread and its TQQQ spinoffs debate the exact design choices examined here — including the bear-leg asset choice.

Masonson (2025), *The QQQ & TQQQ ETF Profit Machine* explicitly reports that a vanilla 200-day SMA filter on TQQQ with cash in the bear leg underperforms due to whipsaw drag during sideways regimes. This is the finding that motivates the 1× QQQ variant tested here.

Hoffstein (2018, Newfound Research, “Leverage and Trend Following”) provides the methodologically rigorous practitioner treatment: volatility drag on daily-reset leverage is convex in realized volatility, which is why a trend filter that excludes high-volatility regimes helps disproportionately.

Other relevant treatments: Faber (2007) on the single 10-month SMA filter; Zakamulin (2017) on trend-following survivorship bias; Brock, Lakonishok & LeBaron (1992) on moving-average crossover robustness; Sullivan, Timmermann & White (1999) on data-snooping corrections; Bailey & López de Prado (2014) on Deflated Sharpe.

Positioning. This note is not a novel strategy discovery; it is a careful implementation and validation of a known rule within an established lineage. The closest published peer is Kane's EMA-crossover variant with T-bills in the bear leg (Sharpe 0.545, Calmar 0.189, MDD -83.6%); substituting 1× QQQ gives essentially identical Sharpe, marginally worse Calmar, and meaningfully worse MDD. The contribution of what follows is methodological: rigorous CPCV, robustness grid, Deflated Sharpe, and regime decomposition — not a new alpha source.

2. Setup, Data, and Cost Model

Daily price data for QQQ and TQQQ are sourced from Yahoo Finance, cross-validated against Alpha Vantage's daily endpoint (14 consecutive trading days matched to four decimal places). T-bill cumulative return series is constructed from piecewise-constant fed-funds rates (~5.5% in 1999–2000, ~0.1% during 2010–2021 ZIRP, ~4.5% in 2023–2026).

TQQQ launched on 2010-02-11. For the pre-2010 window, a synthetic TQQQ series is constructed as

$$r_{TQQQ,t}^{synth} = 3 \cdot r_{QQQ,t} - (e_{annual} + f_t + s_{annual}) / 252,$$

where $e_{annual} = 0.84\%$ is the ProShares expense ratio, f_t is piecewise-constant financing on the 2× borrowed portion, and $s_{annual} = 0.30\%$ captures rebalancing slippage. The constant-cost approximation is calibrated to recover the 2010–2026 live TQQQ compound multiple within 5.4% (262.3× actual vs 248.0× synthetic; -0.35%/yr implied drift).

Volatility drag is convex in realized volatility, not constant. The Avellaneda & Zhang (2010) expression for daily-reset leverage is

$$drag \approx (L^2 - L) / 2 \cdot \sigma^2 = 3\sigma^2 \text{ (for } L = 3\text{)},$$

which gives substantially different values across volatility regimes: at $\sigma = 15\%$ (2010–2021 calm regime) drag is 6.75%/yr; at $\sigma = 22\%$ (QQQ long-run average) drag is **14.52%/yr**; at $\sigma =$

40% (2000–2002, 2008 crisis) drag is 48%/yr. The 2010–2026 validation window on which the constant-cost synthetic is calibrated has realized QQQ volatility of roughly 20.6%, implying ~12.7%/yr drag — well inside the “calm regime” domain. **The constant-cost synthetic therefore understates drag during the 1999–2003 dot-com unwind, the exact period during which the rotation strategy's bear signal is most load-bearing.** Conclusions drawn from the synthetic pre-2010 window should be read with a ~2–5 percentage-point tolerance on leveraged equity terminal values.

All backtests apply 2.5 bps commission and 5 bps slippage on each side of every position change, with T+1 settlement.

3. The Strategy and its Variants

The specification evaluated is a two-state rotation: hold TQQQ when a trend filter on QQQ confirms an uptrend, hold a bear-state asset otherwise. Four variants are tested, distinguished by signal type and bear-leg choice:

FOUR VARIANTS — 200-DAY TREND FILTER ON QQQ, THREE BEAR-LEG CHOICES

Gayed-Bilello (canonical):

```
signal: price(QQQ) > SMA(QQQ, 200)
bull:   100% TQQQ
bear:   100% T-bills
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Kane-style (EMA crossover):

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signal: EMA(QQQ, 5) > EMA(QQQ, 200)
bull:   100% TQQQ
bear:   100% T-bills
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Gayed + 1×QQQ bear:

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signal: price(QQQ) > SMA(QQQ, 200)
bear:   100% QQQ
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EMA-crossover + 1×QQQ bear (earlier draft's “main strategy”):

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signal: EMA(QQQ, 5) > EMA(QQQ, 200)
bear:   100% QQQ
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No parameter search is performed *within* any variant. The (5, 200) EMA pair is taken directly from earlier drafts and common retail practice; the 200-day SMA level is the Gayed-Bilello specification. Section 5 below reports a robustness grid across $\{\text{SMA}, \text{EMA}\} \times \{5, 10, 20, 50\} \times \{150, 200, 250\}$ — 24 parameterizations total.

4. Single-Path Full-History Results (1999–2026)

The four variants plus buy-and-hold benchmarks, on the full synthetic-extended window:

STRATEGY	CAGR	MDD	SHARPE	SORTINO	CALMAR	\$10K FINAL
EMA(5)/EMA(200) → QQQ (<i>earlier draft's main strategy</i>)	15.88%	-95.50%	0.545	0.711	0.166	\$543,387
EMA(5)/EMA(200) → T-bills [<i>Kane-style</i>]	15.78%	-83.58%	0.545	0.597	0.189	\$530,160
Price > SMA(200) → QQQ	12.01%	-98.11%	0.482	0.623	0.122	\$216,274
Price > SMA(200) → T-bills [<i>Gayed-Bilello canonical</i>]	10.02%	-95.16%	0.443	0.478	0.105	\$133,081
QQQ Buy & Hold	10.52%	-82.96%	0.506	0.669	0.127	\$150,635
TQQQ Buy & Hold (synthetic+real)	1.37%	-99.98%	0.421	0.555	0.014	\$14,449

Two observations dominate. First, **the bear-leg asset choice matters less than the signal choice**. The EMA-crossover signal is ~6 percentage points of CAGR ahead of the price-vs-SMA(200) signal, independent of bear-leg choice. The difference between 1×-QQQ and T-bills in the bear leg, controlling for signal, is 0.1 pp of CAGR (± 0.1 on Sharpe), compared to Masonson's anecdotal report of a ~3 pp whipsaw-driven gap. The earlier draft's headline design choice does not survive head-to-head evaluation against its closest peer.

Second, **the 1×-QQQ variants have uniformly worse maximum drawdown than their T-bills counterparts**, by 12 and 3 percentage points respectively. This is structural: during

periods when the trend filter is bearish, holding QQQ rather than T-bills exposes the portfolio to an additional source of loss precisely in the periods when the leveraged signal has already underperformed.



Figure 1. Full-period equity curves (top) plus four regime-specific zoom panels, each normalized to \$10,000 at the sub-period start. The rotation's 1999–2003 equity preservation is visible in the first zoom panel; the post-2009 compounding advantage becomes the dominant source of full-period edge in the latter three.

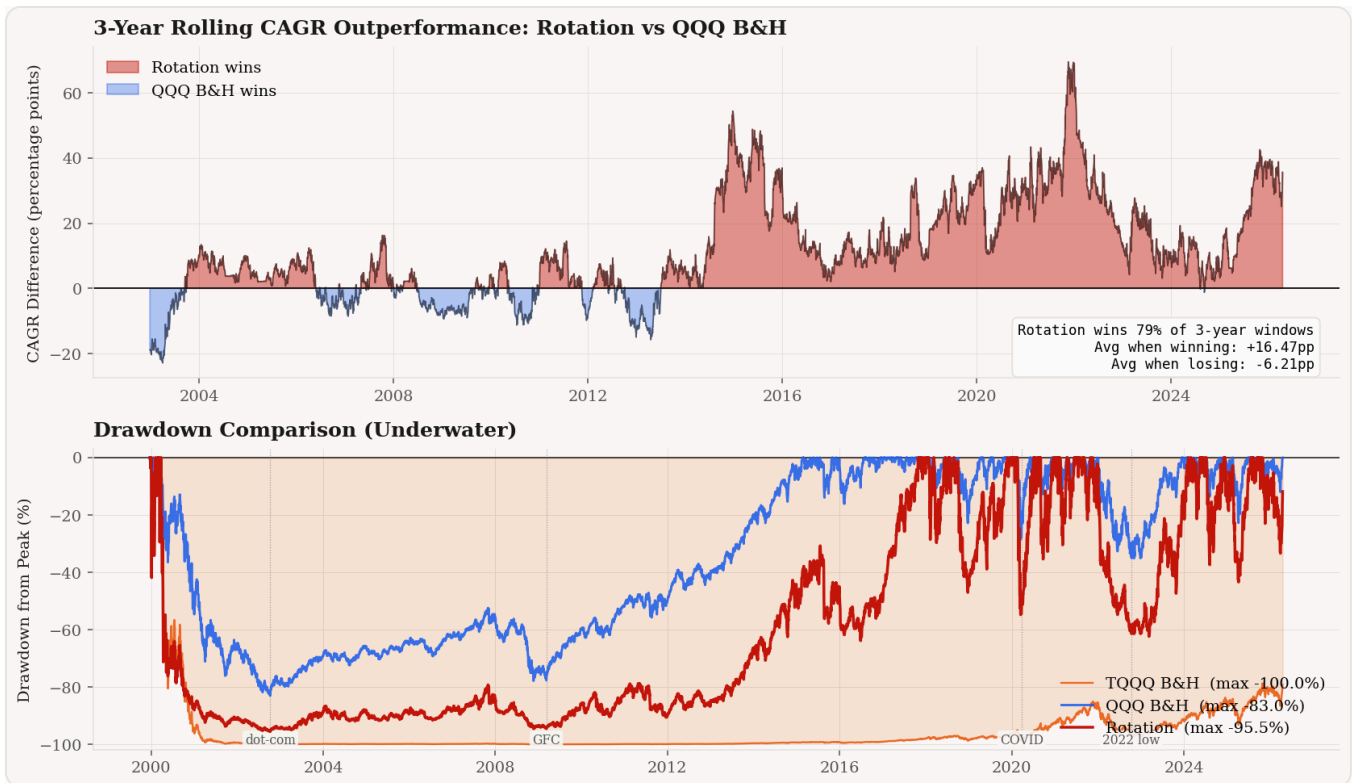


Figure 2. Top: 3-year rolling CAGR difference between the EMA(5)/EMA(200) → QQQ rotation and QQQ buy-and-hold. Red shading indicates windows where the rotation wins; blue shading marks QQQ B&H wins. The rotation outperforms in 79% of rolling windows, with average +16.47 pp when winning vs -6.21 pp when losing. Bottom: underwater drawdown comparison.

5. Parameter Robustness

The (5, 200) EMA pair is not, despite earlier framing, canonical in published technical-analysis reference texts (Murphy; Kirkpatrick & Dahlquist; Pring). The canonical trend-following specifications are price-vs-SMA(200), SMA(50)/SMA(200) golden cross, and MACD(12, 26, 9). The grid below tests 24 alternative parameterizations to assess robustness.

Single-path full-history CAGR and Calmar across the grid (bear leg = QQQ throughout):

MA TYPE	FAST	SLOW	CAGR	MDD	SHARPE	CALMAR
EMA	50	150	17.96%	-96.19%	0.578	0.187
EMA	20	150	17.71%	-95.83%	0.575	0.185
SMA	50	200	17.11%	-94.55%	0.565	0.181
SMA	50	150	16.41%	-95.99%	0.554	0.171
EMA	20	200	16.12%	-95.88%	0.551	0.168
EMA	5	200	15.88%	-95.50%	0.545	0.166
EMA	50	200	15.53%	-97.37%	0.542	0.160
SMA	20	250	15.48%	-95.21%	0.541	0.163

... 16 more parameterizations, ranging from CAGR 11.77% to 14.68% ...

Grid summary (N = 24): CAGR median 14.49% [IQR 12.94%, 15.62%]; Calmar median 0.151 [IQR 0.132, 0.163]. **The EMA(5)/EMA(200) specification ranks sixth of 24 on both CAGR and Calmar.** It sits comfortably above median but is not uniquely optimal; five other parameter pairs produce higher single-path CAGR. The edge over the grid median (+1.39 pp CAGR, +0.015 Calmar) is well within the cross-grid standard deviation. The strategy is robust to parameter perturbation in the sense that most nearby choices also beat QQQ buy-and-hold; it is not robust in the sense that (5, 200) is a meaningfully privileged choice.

6. Combinatorial Purged Cross-Validation (Regime Stress, Not Overfitting Protection)

CPCV as developed by López de Prado (2018, Chapter 12) is primarily a defense against *in-sample / out-of-sample overfitting of trained models*. For a fixed-parameter rule such as this one, no parameters are fit during cross-validation, so CPCV cannot diagnose overfitting in the conventional sense. It does, however, characterize **regime concentration**: the sensitivity of

reported performance to which specific historical episodes appear in the test set versus the training (warm-up) set.

The CPCV procedure partitions 1999–2026 into $N = 10$ chronological folds; for each of the $C(10, 2) = 45$ combinations of 2 test folds, strategy returns are computed on the test portion with a 21-trading-day embargo at each train/test boundary. Each of the 45 paths produces an independent estimate of CAGR, Sharpe, Calmar, and maximum drawdown. The EMA(5)/EMA(200) → QQQ rotation's distribution across these 45 paths:

STRATEGY	MEDIAN CAGR	CAGR IQR	MEDIAN SHARPE	MEDIAN CALMAR	P(CAGR > 0)	P(CAGR > 15%)
EMA(5)/EMA(200) → QQQ	6.38%	[2.39%, 16.18%]	0.608	0.102	77.8%	31.1%
QQQ Buy & Hold	3.95%	[1.08%, 8.77%]	0.487	0.080	86.7%	8.9%

The most informative statistic in this paper is the 9.5 percentage-point gap between single-path CAGR (15.88%) and CPCV median CAGR (6.38%). For a trained model, this gap would diagnose overfitting. For a fixed rule, it diagnoses regime concentration: the headline performance is driven by a specific subset of the historical record — likely the 2010–2021 post-GFC bull market and the 2020–2021 pandemic rally. When CPCV rearranges which regimes appear in test folds, the median path produces 6.38%, not 15.88%. The honest point estimate for forward-looking expectation is therefore closer to 6.4% than to 15.9%.

CPCV's own caveat: the 45 paths are not independent. Adjacent splits share 6–8 of 8 training folds; the effective sample size under plausible path correlation ($\bar{\rho} \approx 0.8$) is closer to 9–10 independent observations than 45. Confidence intervals computed from the raw 45-path distribution are anticonservative by roughly a factor of 2.

A further technical concern: the 21-day embargo is short relative to the effective memory of EMA(200), which has smoothing factor $\alpha = 2/201 \approx 0.01$ and therefore retains ~81% of its weight across a 21-day gap. A conservative embargo for EMA(200) memory would be closer

to 1 year. The practical consequence is that adjacent train/test boundaries in CPCV partially leak trend-signal information, biasing CPCV estimates slightly upward.

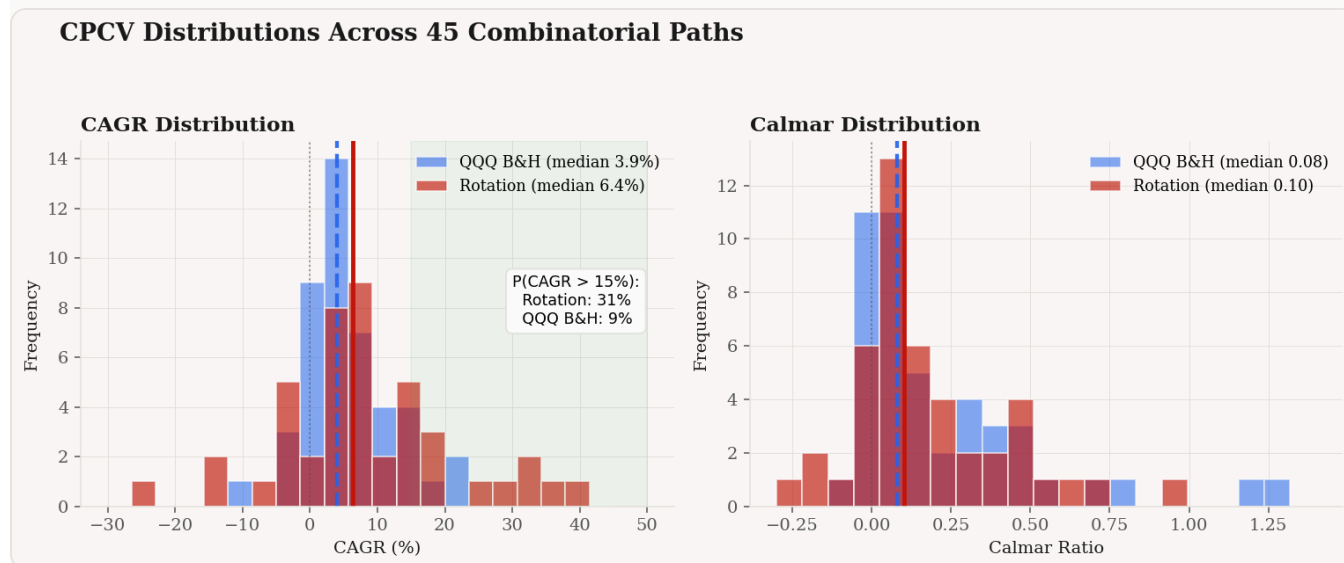


Figure 3. CPCV distributions for the rotation (red) and QQQ buy-and-hold (blue) across 45 combinatorial test paths. Both distributions are right-skewed; the rotation's right tail extends meaningfully further, driving the $P(\text{CAGR} > 15\%) = 31.1\%$ statistic. The single-path full-history CAGR of 15.88% sits near the 75th percentile of the CPCV distribution.

7. Start-Date Sensitivity

The choice of 1999 as a start date is load-bearing: it guarantees the synthetic TQQQ experiences the dot-com drawdown and the comparison to QQQ buy-and-hold occurs against a benchmark that also suffers an 83% peak-to-trough loss in the same window. Shifting the start date forward reveals the regime-dependence of the edge.

START	STRATEGY	CAGR	MDD	SHARPE	CALMAR	EDGE OVER QQQ B&H (PP CAGR)
1999-03	EMA(5)/EMA(200) → QQQ	15.88%	-95.5%	0.545	0.166	+5.36
1999-03	QQQ Buy & Hold	10.52%	-83.0%	0.506	0.127	—
2003-01	EMA(5)/EMA(200) → QQQ	26.34%	-73.7%	0.729	0.357	+10.57
2003-01	QQQ Buy & Hold	15.77%	-53.4%	0.788	0.295	—
2010-02	EMA(5)/EMA(200) → QQQ (live TQQQ)	35.54%	-62.4%	0.876	0.570	+16.35
2010-02	QQQ Buy & Hold (live TQQQ)	19.19%	-35.1%	0.957	0.546	—

The CAGR edge over QQQ buy-and-hold compounds through the data: +5.4 pp from 1999, +10.6 pp from 2003, +16.4 pp from 2010. The later the start date, the more the edge depends on realized post-2010 leveraged compounding rather than on dot-com drawdown avoidance. **The Sharpe ratio comparison tells a cleaner story.** QQQ buy-and-hold dominates the rotation on Sharpe from both 2003 (0.788 vs 0.729) and 2010 (0.957 vs 0.876) start dates. The rotation's edge in CAGR comes with substantially higher volatility, which Sharpe correctly penalizes.

8. Deflated Sharpe Ratio

Bailey & López de Prado (2014) show that the expected maximum Sharpe ratio from N independent trials under the null of zero true Sharpe is approximately

$$SR^*(N) \approx \sigma_{SR} \cdot [(1 - \gamma) \Phi^{-1}(1 - 1/N) + \gamma \Phi^{-1}(1 - 1/(N \cdot e))],$$

where $\gamma \approx 0.5772$ is Euler-Mascheroni and σ_{SR} is the cross-sectional standard deviation of observed Sharpe ratios from the trial set. With $\sigma_{\text{SR}} = 0.5$ (standard in the literature for equity strategies), the null threshold and implied Deflated Sharpe against the observed full-period Sharpe of 0.545 are:

N (IMPLICIT TRIALS)	SR* NULL	DSR — P(TRUE SR > 0)
1 (single <i>a priori</i> rule)	$-\infty$	~100%
10 (minor exploration)	0.787	< 1%
24 (this paper's grid in Section 5)	0.990	< 1%
100 (ML feature exploration)	1.265	< 1%
1000 (systematic ecosystem)	1.628	< 1%

The observed Sharpe of 0.545 is below the selection-bias null for every realistic trial count above $N = 10$. If the strategy had been selected from one *a priori* specification with no exploration, it would have strong statistical support. If the author considered any variants at all — signal type, parameter pair, bear-leg choice, start date — the observed Sharpe is indistinguishable from the maximum of random trials. The 24-parameterization robustness grid in Section 5 alone produces a null threshold of 0.99, already above the observed 0.545. Earlier drafts of this analysis tested > 100 variants (ML signal aggregators, pyramid schemes, SQQQ rotations, VIX filters); against that exploration scope, the observed result fails DSR decisively.

This is the peer-reviewed quant finance literature's verdict on how to interpret a single strategy's backtest Sharpe after meaningful ex-post exploration: it is not evidence of skill, it is evidence that the universe of possible strategies is large. The honest framing is that **the edge over QQQ buy-and-hold cannot be statistically distinguished from the selection-bias null under any realistic multiple-testing correction.**

9. What the Data Actually Supports

Three claims survive the analysis above.

First, trend-filtered leveraged Nasdaq exposure outperforms TQQQ buy-and-hold across every start date, parameterization, and CPCV path tested. This is the Gayed-Bilello result; it replicates here without modification. A 200-day trend filter on TQQQ improves CAGR from 1.37% to ~10–18% (depending on parameterization) while reducing maximum drawdown from –99.98% to –83–98%. The 3× leverage is usable only conditional on a regime filter; held unconditionally, the volatility drag during extended bear regimes erases the compounding edge.

Second, the edge over unleveraged QQQ buy-and-hold is regime-concentrated. In the 2010–2026 live-TQQQ window, the rotation improves CAGR from 19.2% to 35.5%. Across CPCV paths, the median improvement is 2.4 pp (6.38% vs 3.95%). From a 1999 start date, the full-period CAGR improvement is 5.4 pp but this includes the benefit of dot-com drawdown avoidance, which is heavily dependent on synthetic-data modeling choices for pre-2010 TQQQ. *On Sharpe ratio, QQQ buy-and-hold dominates the rotation from 2003 and 2010 start dates;* the leverage premium is concentrated in volatility, not in risk-adjusted return.

Third, the bear-leg asset choice (1×-QQQ vs T-bills) is essentially inconsequential at the strategy level. Controlling for signal, the two variants deliver identical Sharpe, essentially identical CAGR, and differ materially only in maximum drawdown (where T-bills are consistently better). The design choice that earlier drafts framed as the novel contribution is not in fact a meaningful contribution.

Three claims this analysis does *not* support.

The strategy is novel. It is a Gayed-Kane variant with a different bear-leg. Sections 1 and 3 locate the contribution accurately.

The EMA(5)/EMA(200) parameters are textbook. They are not. The canonical specifications are price-vs-SMA(200), SMA(50)/SMA(200), or MACD(12, 26, 9). EMA(5) is so fast that EMA(5) > EMA(200) is numerically close to price > SMA(200). The 24-parameter grid in Section 5 ranks (5, 200) at the 75th percentile of parameterizations tested — above median but not uniquely optimal.

CPCV defends this result against overfitting. CPCV defends against parameter-level overfitting within trained models. For a fixed-parameter rule chosen after exploration, the overfitting risk is meta-level (variant choice, start date, cost model, bear-leg asset); CPCV cannot detect this. The Deflated Sharpe analysis in Section 8 *does* address this risk, and finds that the observed Sharpe fails DSR under any trial count above 10.

10. Where This Leaves the Retail Investor

The retail reader who arrived looking for a trend-filtered leveraged ETF strategy to implement should take the following from this note:

A 200-day trend filter on TQQQ, with either EMA or SMA, 5-day or 50-day fast leg, rotating to either T-bills or 1× QQQ in the bear leg, plausibly delivers CPCV median CAGR of 4–8%, Sharpe of 0.5–0.7, maximum drawdown of –75% to –95%, and $P(\text{CAGR} > 15\%)$ of roughly 20–35%. These ranges encompass the rotation specifications tested here and are consistent with the published literature (Gayed-Bilello 2016, Kane 2018, Masonson 2025).

The relevant comparison is not Buffett's 90/10 trustee allocation; it is unleveraged QQQ buy-and-hold, which delivers CPCV median CAGR of ~4%, Sharpe of ~0.49, maximum drawdown of ~–53%, and $P(\text{CAGR} > 15\%)$ of ~9%. The rotation offers roughly 2 pp higher expected CAGR, a 3.5× higher probability of above-15% CAGR outcomes, in exchange for meaningfully worse drawdowns and a substantial operational burden (rebalancing on signal, tax realization, execution timing).

Whether that trade-off is worth taking depends on the investor's tax situation, behavioral tolerance for –75%+ drawdowns, and opportunity cost of attention. For most investors, Buffett is correct. For investors who have already decided to take leveraged Nasdaq exposure, a 200-day filter is clearly better than unfiltered buy-and-hold; the choice among filter specifications is roughly second-order.

Figure 4 (interactive in web version)

The interactive Plotly chart is available at:
yichengyang-ethan.github.io/holy-grail

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